

1. Project Folder Structure (Vite + React)

`src/` (Main Source Folder)

This is the core working directory where all React code is written.

- **`main.jsx`**
Entry point of the application. It connects React with the HTML file and renders the root component.
- **`App.jsx`**
Root component of the application. It manages and combines all other components.
- **`assets/`**
Stores static files such as images, icons, and fonts.
- **`components/` (or custom folder like `myComponents/`)**
Contains reusable UI components (e.g., Home, Student).
- **`App.css`**
Styles specific to the App component.
- **`index.css`**
Global styles applied across the entire application.

Root Directory Files

- **`index.html`**
The single HTML file used in the application (Single Page Application concept). Contains a root `<div>` where React renders the UI.
- **`package.json`**
Configuration file that includes dependencies, scripts, and project metadata.
- **`package-lock.json`**
Automatically generated file that locks dependency versions.
- **`vite.config.js`**
Configuration file for Vite.
- **`.gitignore`**
Specifies files/folders to ignore in version control (e.g., `node_modules`).
- **`eslint.config.js`**
Configuration for code linting and quality checks.

2. `npm create vite@latest` Explanation

`npm` (Node Package Manager)

- A tool used to install, manage, and run JavaScript packages.
- Comes with Node.js.
- Used for installing dependencies and running scripts.

Vite

- A modern build tool and development server.
- Provides fast startup and instant updates using Hot Module Replacement (HMR).
- Faster than older tools like Create React App.

@latest

- Installs the most recent version of Vite.
- Ensures you are using updated features and fixes.

3. `npm install` Concept

- Installs all dependencies listed in `package.json`.
- Creates a `node_modules` folder containing installed packages.
- Must be run after creating or cloning a project.

Purpose:

To prepare the project environment so it can run properly.

4. `npm run dev` Concept

- Starts the development server.
- Runs the project locally (e.g., `http://localhost:5173`).
- Enables live updates without reloading the page.

Where does `npm run dev` come from?

It is defined inside `package.json`:

```
"scripts": {  
  "dev": "vite",  
  "build": "vite build",  
  "preview": "vite preview"  
}
```

- `dev` is the script name
- `vite` is the command executed

5. Changing `npm run dev` to Another

Name You can rename the script in `package.json`.

Example:

```
"scripts": {  
  "start": "vite",  
  "build": "vite build"  
}
```

Now run:

```
npm run start
```

Key Rule

- Script names are user-defined.
- Command format:

```
npm run <script-name>
```

6. Complete Workflow

```
npm create vite@latest → Create project npm install  
→ Install dependencies npm run dev → Start  
development server
```

Summary

- `npm` manages packages and scripts
- `vite` is a fast build tool and dev server
- `@latest` installs the newest version
- `npm install` sets up dependencies
- `npm run dev` starts the local server
- Scripts can be modified in `package.json`